

Electric Vehicle: Present and Future

Basudeb Kundu, Prohlad Chandra Das, Debashis Kumar Das, Indronill Dutta Nill,

Nafiur Rahman Nirob

23-50856-1 , 23-50922-1 , 23-50953-1 , 23-50974-1 , 23-50991-1

GROUP - 04

Abstract

Electric Vehicles (EVs) are revolutionizing the transportation industry by offering a cleaner and more sustainable alternative to traditional internal combustion engine vehicles. With rapid advancements in battery technology, power electronics, and charging infrastructure, EVs are becoming increasingly practical for everyday use. This paper provides a brief overview of the current state of EVs, their future prospects, and highlights how simulation tools like Proteus can be used to model and understand EV systems.

Keywords

Electric Vehicle, Battery Technology, Motor Control, PWM, Sustainability, Automation, Vehicle-to-Grid

I. INTRODUCTION

The transportation sector accounts for a significant share of global greenhouse gas emissions. Electric Vehicles (EVs) have emerged as a promising solution by offering zero tailpipe emissions and higher energy efficiency compared to internal combustion engine (ICE) vehicles. Governments and manufacturers are accelerating adoption through policy incentives, declining battery costs, and expanding charging networks. The lecture used as the

foundation for this paper emphasized how replacing an ICE with an electric motor simplifies the drivetrain, reduces moving parts, and creates opportunities for more intelligent control systems.

Early EVs in the late 19th century were eclipsed by gasoline cars due to limited range and infrastructure. However, improvements in lithium-ion batteries, power electronics, and lightweight materials have renewed interest. Today, EVs are seen not only as cars but as platforms for software-driven innovation. The convergence of clean energy and digitalization is reshaping how mobility is conceived and delivered.

II. BACKGROUND AND LITERATURE

Research on EV technology spans multiple disciplines. Battery development remains central, with lithium-ion cells offering energy densities above 250 Wh/kg, enabling ranges of 300–600 km per charge [1]. Advances in thermal management, battery management systems (BMS), and charging algorithms have improved reliability and lifespan. Motor technology includes permanent magnet synchronous motors (PMSM), induction motors, and switched reluctance designs [2]. Each topology provides trade-offs between

cost, efficiency, and torque characteristics.

Infrastructure studies highlight the importance of widespread fast-charging stations and smart charging coordination [3]. Government incentives and corporate sustainability goals are pushing for fleet electrification in logistics, public transport, and personal vehicles. Research also focuses on second-life use and recycling of batteries to minimize environmental impacts, as well as on developing circular supply chains.

From a systems perspective, integrating EVs with renewable energy resources through vehicle-to-grid (V2G) technology enables demand response and grid stabilization. EV adoption is closely tied to energy policy, urban planning, and consumer behavior, making it a truly interdisciplinary field.

III. DESIGN PRINCIPLES OF EVs

A laboratory prototype discussed in the lecture illustrates EV subsystems and their interconnections. Two direct current (DC) motors are used to drive the left and right wheels, while an Arduino Mega 2560 microcontroller manages inputs and outputs. Manual navigation is implemented using push buttons for forward, reverse, braking, and right-turn operations. Pulse-width modulation (PWM) controls the speed of the motors by adjusting the duty cycle of voltage applied to the motor driver, allowing smooth acceleration.

The power supply is routed through a key switch or push button, with optional relays to isolate high-current circuits.

This approach enhances safety by preventing logic-level circuits from being exposed to motor currents. Relays also facilitate modularity, allowing additional subsystems such as lighting or auxiliary actuators to be connected without redesigning the entire control circuit.

Another key insight is the benefit of independent control of each wheel. While a basic prototype may couple both wheels on a single shaft, keeping them separate allows for future implementation of advanced features such as skid steering, autonomous navigation, and even robotic transformations. State variables in the control software track operating modes (e.g., forward, reverse, brake), ensuring clarity and fault tolerance during operation.

IV. CURRENT STATE OF THE ART

Modern EVs have evolved from simple electric drivetrains to sophisticated mechatronic systems. Battery packs are composed of thousands of cells arranged in modules, monitored by BMS for voltage, temperature, and charge balancing. Thermal management uses liquid cooling or phase-change materials to maintain cell health. Regenerative braking recovers kinetic energy, improving efficiency by up to 30%. Fast-charging networks delivering up to 350 kW allow many EVs to recharge 80% of their capacity in less than 30 minutes.

Beyond passenger cars, electrification is reshaping buses, delivery vans, two-wheelers, and construction equipment. Fleet operators benefit from lower maintenance costs, thanks to the absence of oil changes and fewer

mechanical parts. Automakers are increasingly adopting 'skateboard' platforms that integrate batteries, motors, and suspension into a flat chassis, simplifying manufacturing and enabling diverse body styles.

Software-defined vehicles are a parallel trend. Over-the-air updates let manufacturers improve performance, add features, or fix bugs post-sale. Cybersecurity, sensor fusion, and data privacy are gaining prominence as EVs become more connected. Companies are also investing in second-use applications for EV batteries, such as stationary energy storage, extending their lifecycle.

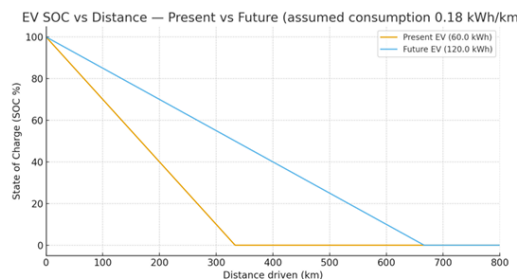
V. Simple Simulation:

EV Battery Range Comparison: To understand EV improvements, here's a simple simulation. - Present EV: 60 kWh battery - Future EV: 120 kWh battery (double energy) - Energy use: ~0.18 kWh/km (typical for mid-size EVs)

Metric	Battery Capacity (kWh)	Estimated Range (km)	Energy Use (kWh/km)
Present EV	60	~333	0.18
Future EV	120	~666	0.18

Figure 1: Simulation showing SOC (%) vs Distance for Present EV and Future EV

Figure 1: Simulation showing SOC (%) vs Distance for Present EV and Future EV



VI. FUTURE DIRECTIONS

The next generation of EVs promises significant breakthroughs in energy storage, intelligence, and sustainability. Solid-state batteries aim to eliminate liquid electrolytes, improving safety and energy density, while sodium-ion and lithium-sulfur chemistries offer alternatives that reduce reliance on scarce materials. Research into ultrafast charging could make refueling an EV nearly as quick as pumping gasoline.

Autonomous driving is another transformative trend. Electric drivetrains pair naturally with sensors, cameras, and artificial intelligence processors, forming the backbone of self-driving taxis, delivery robots, and personal mobility pods. Independent wheel control, as noted in the prototype, is an enabling technology for precise maneuvering in robotic platforms.

Vehicle-to-grid (V2G) services will allow EVs to act as distributed energy resources, storing excess renewable power and feeding it back during demand peaks. This bi-directional relationship will blur the boundary between transport and energy sectors. Policymakers and industry leaders are expected to strengthen standards for recycling and ethical sourcing of materials, ensuring EVs contribute to a truly sustainable ecosystem.

VII. CONCLUSION

Electric vehicles symbolize a convergence of clean energy, advanced electronics, and digital innovation. Their current success stems from rapid technological progress, declining costs,

and supportive regulations. Looking forward, breakthroughs in batteries, automation, and sustainable manufacturing will expand EV capabilities beyond mere transportation to multifunctional, intelligent platforms. For engineers and researchers, EVs offer an open field for creativity—from modular architectures and sensor-rich autonomy to integration with renewable energy systems.

To realize this potential, stakeholders must address challenges such as battery recycling, equitable access to charging, and cybersecurity. Collaboration across disciplines will ensure that EVs continue to evolve as safe, efficient, and environmentally responsible components of future mobility and power networks.

ACKNOWLEDGMENT

The author(s) would like to acknowledge the contributions of educators and industry experts who provided foundational insights into EV design, including the laboratory demonstrations of PWM speed control and motor driver integration.

REFERENCES

- [1] BloombergNEF, “Electric Vehicle Outlook 2024,” BloombergNEF, 2024.
- [2] C. C. Chan, “The State of the Art of Electric, Hybrid, and Fuel Cell Vehicles,” *Proceedings of the IEEE*, vol. 95, no. 4, pp. 704–718, 2007.
- [3] IEA, “Global EV Outlook 2024,” International Energy Agency, 2024.
- [4] J. Larminie and J. Lowry, *Electric Vehicle Technology Explained*, 2nd ed., Wiley, 2012.
- [5] M. Ehsani, Y. Gao, and S. Longo, *Modern Electric, Hybrid Electric, and Fuel Cell Vehicles*, CRC Press, 2018.